ACT I · THE SETTING

Ready by March: Planning for Both Quiet and Extreme Iceberg Years

How Newfoundland's 4 offshore oil rigs spot, tow, and dodge icebergs together

PRESENTED BY TEAM 5

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ACT I · THE CHARACTERS

Two Rig Types: Stand Fast or Sail Away?

86.85%

of Daily Oil ($14.4M/day)

Fixed Concrete Rigs (GBS):
Hibernia • Hebron
Anchored to seafloor: cannot move
Defense: spot early & tow icebergs away

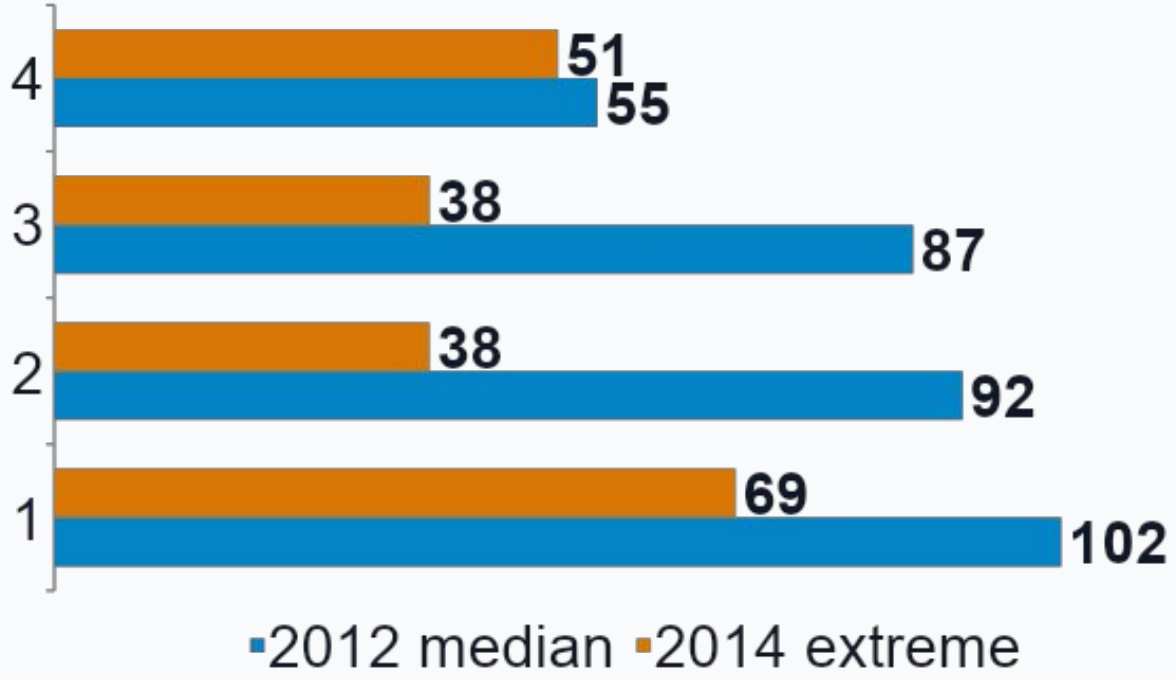
13.15%

of Daily Oil ($2.2M/day)

Floating Ship Rigs (FPSO):
Terra Nova • SeaRose
Can unhook from wells & sail away
Defense: timed countdown to unhook safely

ACT II · PLOT TWIST #1

More Icebergs in the Ocean ≠ More Near the Rigs



154

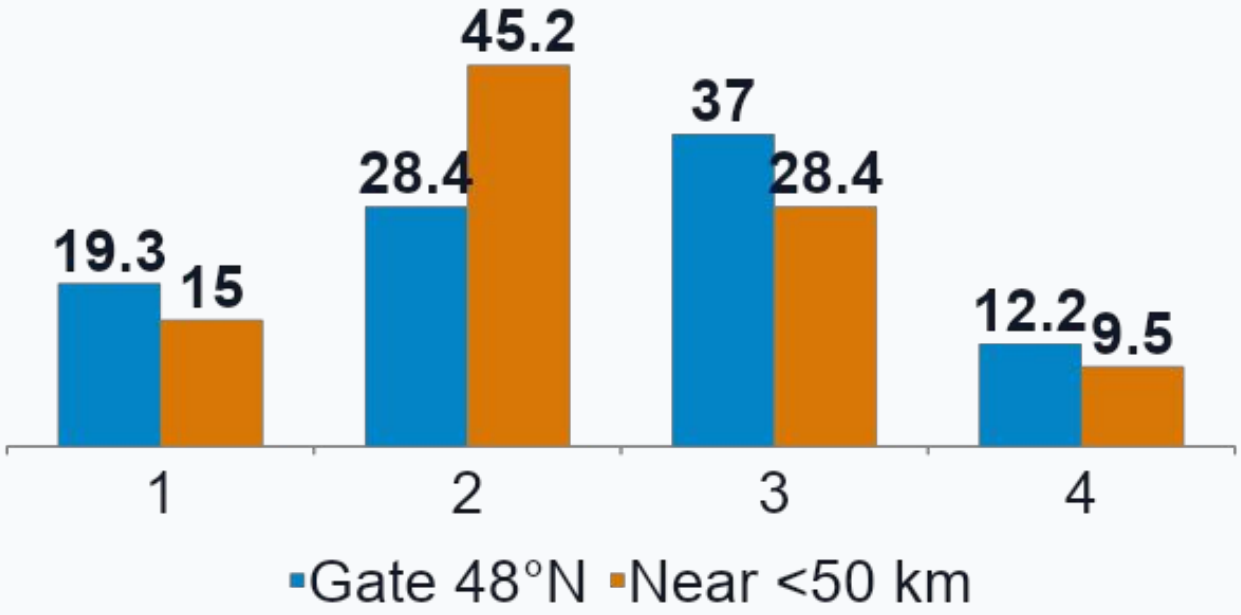
close calls (<50 km) in 2012
('Average' year: 499 in region)

111

close calls (<50 km) in 2014
('Extreme' year: 1,546 in region)

ACT II · SHARED DANGER

Danger Near the Rigs Peaks in April, One Month Before the Ocean Peak



581

times the same iceberg threatened

Hebron & Terra Nova on the same day

(Just 8.2 km apart: Terra Nova can sail away, Hebron cannot)

ACT III · THE CLIMAX · DAY 4

05:20 AM: What Happens When a Tow Rope Slips?

02:30	MONITOR	Iceberg tracked miles out
03:40	MANAGE	Boat readies tow rope
04:32	PREP	Wells shut off; ready to unhook in 15 min
05:20	SLIP	Tow rope slips, 10 min out
05:30	CLEAR	Drifts past safely

Lesson: Prepare to unhook while towing—never wait until a tow rope fails

ACT III · RULES OF ENGAGEMENT

1 in 5 Tow Ropes Slip—Who Makes the Final Call?

22%

of tow ropes slip off
86% deflect the iceberg
(99% avoid downtime)

Rig Boss

Offshore Manager (OIM)
makes sole call to unhook;
onshore team only advises

All 4 Rigs

Share one live iceberg map
and support ships;
each rig
sets its own safety zones

ACT III · PLOT TWIST #2

41% of Nearby Icebergs Had Zero Prior Tracking in the Log

1,555
unique icebergs entered the 50 km zone near rigs

38.3 hrs
typical tracking time from first record to closest point

40.6%
the very first record we have is already their closest point.

ACT III · PUSHING BACK THE HORIZON

Why All-Weather Satellites Matter—And What to Share

13.6%

spotted by satellite in 2021
(Near zero before 2017;
sees through fog & clouds)

1 Keep on each rig (In-House)
24/7 radar, lookouts & manager's call to unhook

2 Rent for spring ice season
Spotter planes, radar satellites & towboats

3 Share across all 4 rigs
One shared iceberg map, joint flights & boat pool

ACT IV · THE STAKES

Leaving Station vs. Staying Put

Field	Can leave?	Gross / day
Hebron	No	$9.5M
Hibernia	No	$4.9M
Terra Nova	Yes	$1.5M
SeaRose (2 fields)	Yes	$0.66M
Basin total		**$16.6M**

ACT IV · THE PROOF · 2014

1,513 Icebergs. Production Held.

225.8K

spring 2014 avg, bbl/d

211.5K

rest-of-year avg, bbl/d

153.6K

ice-free August: planned turnarounds

ACT V · THE RESOLUTION

March 31: Choose the Iceberg Response

Use the number of icebergs crossing 48°N by March 31 to plan spring resources.

QUIET YEAR

Fewer than 10 icebergs

Share one tracking map and one support vessel.

UNCERTAIN YEAR

10–199 icebergs

Keep surveillance flights and tow vessels on standby.

EXTREME YEAR

200+ icebergs

Coordinate radar-satellite monitoring and flights; pool support vessels.

Share tracking and vessels.
Each offshore manager keeps the final say on disconnecting.

NL OFFSHORE ICE MANAGEMENT WORKING GROUP

Questions & Discussion

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